

A Combined Particle/Finite-Difference Scheme for Common-Noise Mean-Field Game FBSDEs: Convergence Rate and CFL Condition

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Abstract

We develop and analyse a fully discrete numerical scheme for the conditional McKean–Vlasov forward-backward stochastic differential equation (FBSDE) system characterising mean-field game (MFG) equilibria with common noise. The scheme combines an Euler–Maruyama discretisation of the forward state and backward adjoint processes with a particle (or finite-difference) approximation of the stochastic Fokker–Planck equation governing the population measure. We prove that the resulting discrete equilibrium converges to the true equilibrium at rate $O(\Delta t^{1/2} + \Delta x)$ in the Wasserstein-2 distance, and derive a CFL-type stability condition for the coupled system that accounts separately for parabolic diffusion, stochastic transport from the common noise, and the explicit feedback coupling between the forward and backward components. We validate the scheme’s strong-order convergence on a tractable linear-quadratic benchmark and discuss the practical particle-count requirements implied by the CFL condition in higher dimensions.

Keywords: mean-field games, common noise, numerical methods, particle methods, CFL condition, McKean–Vlasov FBSDEs

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1 Introduction

The probabilistic characterisation of mean-field game (MFG) equilibria with common noise as solutions to a conditional McKean–Vlasov forward-backward stochastic differential equation (FBSDE) system is now well established (Carmona and Delarue, 2018; Ahuja et al., 2019), and a companion paper establishes an explicit, parameter-dependent contraction estimate for the associated mean-field fixed-point map (Monyimba, 2026). Computing the equilibrium in practice, however, requires discretising three coupled components simultaneously: the forward state SDE, the backward adjoint BSDE, and the stochastic partial differential equation (SPDE) governing the evolution of the population measure under the common shock. The third component is the distinguishing numerical difficulty relative to the classical (no-common-noise) MFG setting, where the population measure is deterministic and solves an ordinary (not stochastic) Fokker–Planck PDE.

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Contribution. We propose a combined Euler–Maruyama/particle scheme for the full coupled system (Section 3) and prove (Theorem 4.1) that it converges to the true equilibrium measure flow at rate $O(\Delta t^{1/2} + \Delta x)$ in Wasserstein-2 distance, under the same structural assumptions used to establish well-posedness in Monyimba (2026). The proof (Section 5) combines weak-error estimates for the forward/backward components with a strong-error estimate for the stochastic Fokker–Planck discretisation, then closes the loop via the contraction property of the continuous-time fixed-point map. We further derive an explicit CFL-type condition (Section 6) for the coupled system, which – unlike the classical parabolic CFL condition for deterministic Fokker–Planck PDEs – includes an additional constraint from the stochastic transport induced by the common noise.

Numerical validation, and an important caveat. We validate the scheme’s *strong*-order convergence (pathwise L^2 error against a fine-resolution reference path, using common random numbers) on the tractable linear-quadratic (LQ) benchmark in Section 7, confirming the expected rate of 1.0 for this benchmark (additive, state-independent diffusion coefficients give the Milstein correction term a zero coefficient, so Euler–Maruyama already attains its maximal strong order). **We note explicitly that a full numerical verification of the theorem’s *weak* (Wasserstein) rate on the LQ benchmark, using a properly noise-controlled (common-random-number or paired) Monte Carlo design at adequate particle counts, remains to be completed and is not reported here; our own preliminary attempt at a naive (non-paired) weak-error sweep did not exhibit a clean convergence trend at the particle counts we tested, indicating that Monte Carlo sampling noise dominates the discretisation signal at those settings.** We report only the strong-error results, which we have validated repeatedly and which directly confirm the scheme’s stability and the temporal discretisation order; the weak-error LQ-benchmark table given in an earlier preprint version of this material was generated using an erroneous sign in the benchmark’s closed-form Riccati coefficients and is superseded by the corrected system given in Appendix A.

Related work. Chassagneux et al. (2019) establish convergence of a particle method for McKean–Vlasov FBSDEs without common noise. Achdou and Porretta (2016) analyse finite-difference schemes for deterministic MFG PDEs. Lord et al. (2014) provide the general SPDE numerics background underlying our Fokker–Planck discretisation. Our contribution is the extension of the particle/FBSDE convergence theory to the common-noise setting, where the measure-valued component is itself stochastic, together with the explicit CFL condition this introduces.

Organisation. Section 2 recalls the FBSDE system. Section 3 defines the discretisation. Section 4 states the main convergence theorem. Section 5 sketches the proof. Section 6 derives the CFL condition. Section 7 reports the validated strong-error experiment. Section 8 discusses scope and the open weak-error verification.

2 The FBSDE System

We adopt the setting of Monyimba (2026): a representative agent’s state $X_t \in \mathbb{R}^d$ solves $dX_t = b(t, X_t, \mu_t, \alpha_t)dt + \sigma dW_t^i + \sigma_0 dW_t^0$, with $\mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$, optimal control characterised by the stochastic maximum principle via the adjoint BSDE $-dp_t = \partial_x H dt - q_t dW_t^i - r_t dW_t^0$, and the equilibrium measure flow μ^* the unique fixed point of the mean-field map Φ . Under the standing assumptions of Monyimba (2026), μ^* satisfies the (weak form of the) stochastic Fokker–Planck

equation

$$d\langle \mu_t, \phi \rangle = \langle \mu_t, \mathcal{L}_t^* \phi \rangle dt + \sigma_0 \langle \mu_t, \nabla \phi \rangle dW_t^0, \quad \phi \in C_c^\infty(\mathbb{R}^d), \quad (1)$$

where $\mathcal{L}_t^* \phi = -\nabla \cdot (b\phi) + \frac{\sigma^2}{2} \Delta \phi$.

3 The Discrete Scheme

Fix a time grid $t_n = n\Delta t$, $n = 0, \dots, N$, $\Delta t = T/N$. The forward state and backward adjoint are discretised by Euler–Maruyama,

$$X^{n+1} = X^n + b(t_n, X^n, \mu^n, \alpha^n) \Delta t + \sigma \Delta W_n^i + \sigma_0 \Delta W_n^0, \quad (2)$$

$$p^n = \mathbb{E}_n[p^{n+1} + \partial_x H(t_n, X^n, \mu^n, p^n, \alpha^n) \Delta t], \quad (3)$$

with the conditional expectation in (3) approximated by regression on a finite-dimensional basis (Gobet et al., 2005). The measure μ^n is approximated either by the empirical measure of M simulated particles or by an explicit finite-difference scheme on a spatial grid of mesh Δx for the strong form of (1); in the particle case there is no explicit Δx , and the effective resolution scales as $O(M^{-1/d})$.

4 Main Theorem

Theorem 4.1 (Numerical convergence). *Under the standing assumptions of Monyimba (2026) and the additional smoothness condition that b, f, g are of class $C^{2,4}$ with bounded derivatives, there exists $C > 0$, independent of $\Delta t, \Delta x$, such that for all sufficiently small $\Delta t, \Delta x$ satisfying the CFL condition (5),*

$$\sup_{0 \leq n \leq N} \mathbb{E}[W_2(\mu^{\Delta, n}, \mu_{t_n}^*)^2]^{1/2} \leq C(\Delta t^{1/2} + \Delta x). \quad (4)$$

Proof sketch. See Section 5. □

5 Proof Sketch

The proof combines three error contributions and closes the loop via the contraction property of Φ established in Monyimba (2026).

(i) Forward/backward weak error. The Euler–Maruyama weak error for Lipschitz SDEs under the stated smoothness is $O(\Delta t)$ (Kloeden and Platen, 1992, Thm 14.5.2); an analogous $O(\Delta t)$ weak error holds for the BSDE discretisation (Bouchard and Touzi, 2004; Gobet et al., 2005). Both propagate to an $O(\Delta t^{1/2})$ contribution to the Wasserstein distance between discrete and continuous laws via the synchronous-coupling argument of Monyimba (2026).

(ii) Fokker–Planck discretisation error. The explicit finite-difference (or particle) scheme for (1) satisfies, under the CFL condition (5) and by a standard Lax–Richtmyer consistency-plus-stability argument adapted to the stochastic transport term (Lord et al., 2014, Thm 10.7),

$$\sup_n \mathbb{E}[\|\rho^n - \rho_{t_n}\|_{L^2}^2]^{1/2} \leq C(\Delta t^{1/2} + \Delta x).$$

(iii) Closing the fixed-point loop. Let Φ^Δ denote the discrete fixed-point map induced by (2)–(3) together with the Fokker–Planck discretisation. Combining (i)–(ii) gives $\|\Phi^\Delta(\mu) - \Phi(\mu)\|_{\lambda, T} \leq C_{\text{disc}}(\Delta t^{1/2} + \Delta x)$ uniformly over bounded sets of measure flows. Writing $\mu^\Delta = \Phi^\Delta(\mu^\Delta)$

for the discrete fixed point and using the triangle inequality with the contraction factor $\rho < 1$ from Monyimba (2026),

$$\|\mu^\Delta - \mu^*\|_{\lambda, T} \leq C_{\text{disc}}(\Delta t^{1/2} + \Delta x) + \rho \|\mu^\Delta - \mu^*\|_{\lambda, T},$$

and rearranging gives (4) with $C = C_{\text{disc}}/(1 - \rho)$.

6 CFL Condition for the Coupled System

Stability of the finite-difference Fokker–Planck discretisation requires the parabolic condition $\Delta t \leq \Delta x^2/(2\sigma^2)$. The stochastic transport term introduced by the common noise adds the further constraint $\Delta t \leq \Delta x/(2\|\sigma_0\|_F L_b)$ (Lord et al., 2014, Ch. 10). The explicit coupling between the discrete forward state and backward adjoint (via $\partial_x H$ evaluated at the current discrete state) introduces a third constraint, $\Delta t \leq 1/L_H$ with L_H the Lipschitz constant of $\partial_x H$. Collecting these,

$$\Delta t \leq \min \left(\frac{\Delta x^2}{2\sigma^2}, \frac{\Delta x}{2\|\sigma_0\|_F L_b}, \frac{1}{L_H} \right). \quad (5)$$

In the particle method, Δx has no direct analogue; the effective resolution scales as $O(M^{-1/d})$, so (5) translates into a joint requirement on the particle count M and the time step that becomes increasingly demanding as the state dimension d grows – the familiar particle-method curse of dimensionality, here interacting with the common-noise stochastic-transport constraint specifically.

7 Numerical Validation: Strong-Order Convergence

We validate the temporal discretisation order on the one-dimensional linear-quadratic benchmark with $\kappa = 0.5, \lambda = 1.0, \sigma = 0.3, \sigma_0 = 0.2, T = 1$, comparing trajectories at four step sizes against a fine-resolution ($\Delta t = 2^{-10}$) reference path, using common random numbers (CRN) to isolate the discretisation error from Monte Carlo sampling noise. We report the *strong* (pathwise L^2) error, $\mathbb{E}[|X_T^\Delta - X_T^{\text{ref}}|^2]^{1/2}$, averaged over 100 independent trials.

Table 1: Strong-error convergence, LQ-MFG benchmark (CRN, 100 trials)

Δt	Mean error	Std. error	Estimated rate
$2^{-5} = 0.03125$	0.01751	0.01001	–
$2^{-6} = 0.01563$	0.01030	0.00796	–
$2^{-7} = 0.00781$	0.00540	0.00441	–
$2^{-8} = 0.00391$	0.00218	0.00127	0.996

The estimated rate of 0.996 matches the theoretical strong order for additive (state- and measure-independent) diffusion coefficients, for which the Milstein correction term vanishes and Euler–Maruyama already attains strong order 1 rather than the generic order 1/2. **This confirms the temporal discretisation is implemented correctly and is stable, but it validates the *strong-error* behaviour of the forward SDE component in isolation; it does not, by itself, validate the *weak* (Wasserstein, measure-level) rate of Theorem 4.1, which governs the full coupled scheme’s convergence to the equilibrium measure and is the theorem’s actual content.** We regard a properly noise-controlled empirical verification of the weak rate as a required next step before this manuscript is finalised for submission (see Section 8).

8 Discussion

8.1 What is, and is not, empirically confirmed here

The strong-error experiment of Section 7 confirms that the forward discretisation is implemented correctly and behaves as theory predicts for this benchmark’s specific (additive-noise) structure. It does *not* confirm the weak/Wasserstein rate $O(\Delta t^{1/2} + \Delta x)$ of Theorem 4.1, which is a statement about convergence of the *measure*, not a single trajectory, and requires a separate experimental design (e.g., comparing the empirical measure of many particles against the known closed-form Gaussian equilibrium measure via the Wasserstein-2 formula for Gaussians, using either a much larger particle count or a common-random-number / paired-trajectory design to suppress Monte Carlo sampling noise enough to resolve the $O(\Delta t^{1/2})$ trend). A preliminary, non-paired attempt at this experiment did not show a clean trend at the particle counts tested, indicating the sampling-noise floor was not adequately controlled; we have not yet completed a design that resolves this and do not report specific weak-error numbers as a result.

8.2 Practical implications of the CFL condition

Equation (5)’s three terms have different practical bite depending on the application. For low-dimensional financial models ($d \lesssim 5$), the parabolic and coupling terms typically dominate, and standard grid resolutions suffice. For higher-dimensional models, the particle method’s $O(M^{-1/d})$ effective resolution makes the stochastic-transport term increasingly binding, motivating the dimension-reduction techniques (moment closures, low-rank measure approximations) we discuss as future work.

8.3 Limitations

Beyond the open weak-error verification noted above: the theorem requires $C^{2,4}$ smoothness, which excludes models with non-smooth transaction costs or jump-type frictions common in financial applications; the particle method’s curse of dimensionality is not addressed by any variance-reduction technique in the present scheme; and the contraction factor ρ inherited from Monyimba (2026) is itself only an expectation-level (not pathwise) guarantee, which limits the strength of any high-probability (rather than expectation) statement one might hope to extract from Theorem 4.1.

9 Conclusion

We have proposed and analysed a combined particle/finite-difference scheme for common-noise MFG equilibria, proving an $O(\Delta t^{1/2} + \Delta x)$ Wasserstein convergence rate and deriving the CFL condition the coupled system requires. The scheme’s temporal discretisation is validated via a strong-error experiment on a tractable benchmark; a corresponding empirical validation of the theorem’s weak (measure-level) rate remains open and is flagged explicitly as such, rather than asserted without adequate experimental support.

A Corrected Linear-Quadratic Benchmark Coefficients

For the benchmark of Section 7, the closed-loop optimal control is $\alpha_t^* = -(A_t X_t + B_t \bar{\mu}_t)/\lambda$ with (A_t, B_t) solving

$$\dot{A}_t = 2\kappa A_t - A_t^2/\lambda + 2Q, \quad A_T = 2P, \quad (6)$$

$$\dot{B}_t = \kappa B_t + A_t B_t/\lambda - \kappa A_t, \quad B_T = -2P. \quad (7)$$

We flag explicitly that the sign of the cross-coupling term in the B_t equation (here, $+A_t B_t/\lambda$) was found, during preparation of this manuscript, to have been given with the opposite sign in an earlier internal draft; the corrected sign above was verified by an independent empirical optimality check (perturbing the closed-loop control away from its Riccati-implied value and confirming that realised cost strictly increases, as required at a true optimum) before being adopted here. For the stated parameters, $A_t \equiv 2$ for all t (a fixed point of the A -equation), and B_t ranges from approximately 0.20 at $t = 0$ to -2 at $t = T$.

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